MFML – Assignment 2 . 2025ab05055

**Q.** Given $\hat{y} = w_0 + w_1 x_i$

| shop number ($i$) | operating house ($x_i$) | Consumption ($y_i$) |
|---|---|---|
| 1 | 1 | 3 |
| 2 | 2 | 5 |
| 3 | 4 | 9 |
| 4 | 6 | 13 |
| 5 | 8 | 17. |

Loss function $J(w_0, w_1) = \frac{1}{2} \sum_{i=1}^{5} (\hat{y}_i - y_i)^2$

**Sol^n** (a).

Step-1 . prediction errors.

$$\hat{y}_i - y_i = (w_0 + w_1 x_i - y_i)$$

substitute value.

$$J(w_0, w_1) = \frac{1}{2}\Big[(w_0 + w_1 - 3)^2 + (w_0 + 2w_1 - 5)^2 + (w_0 + 4w_1 - 9)^2 + (w_0 + 6w_1 - 13)^2 + (w_0 + 8w_1 - 17)^2\Big]$$

Step-2 Hessian matrix.

for squared error loss. $H = \begin{bmatrix} \sum 1 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$

compute sum:

$\sum 1 = 5$, $\sum x_i = 21$, $\sum x_i^2 = 121$

$$H = \begin{bmatrix} 5 & 21 \\ 21 & 121 \end{bmatrix}$$

Step-3. check convexity

$H_{11} = 5 > 0$

Now find determinant

$$\det(H) = 5(121) - 21 \times 21.$$

$$= 605 - 441 = 164 > 0.$$

Hence Hessian is positive definite

So. $J(w_0, w_1)$ is convex.

(b) Gradient of Loss function

$$\frac{\partial J}{\partial w} = \sum_{i=1}^{5} (\hat{y}_i - y_i)$$

$$\frac{\partial J}{\partial w_1} = \sum_{i=1}^{5} (\hat{y}_i - y_i) x_i$$

(c) Gradient Descent:

initial values $w_0^0 = 2, w_1^0 = 2$

predicted values: $\hat{y} = [4, 6, 10, 14, 18]$

Errors. $[1, 1, 1, 1, 1]$

$$\frac{\partial J}{\partial w_0} = 5, \quad \frac{\partial J}{\partial w_1} = 21$$

Gradient $\nabla J = \begin{bmatrix} 5 \\ 21 \end{bmatrix}$

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(a) C-1 Lets assume constant learning rate $\eta = 0.05$

Iteration 1 $w^1 = w^0 - \eta \nabla J$

$w_0^1 = 2 - 0.05(5) = 1.75$

$w_1^1 = 2 - 0.05(21) = 0.95$

Iteration 2 $w_0^2 = 1.75 - 0.05(5) = 1.75 - 0.25 = 1.5$

$w_1^2 = 0.95 - 0.05(21) = 0.95 - 1.05 = -0.10$

C-2 Lets assume exponentially decaying $\eta = \eta_0 e^{-kt}$

Iteration 1: $\eta_1 = 0.1 e^{-0.4} = 0.067$

$w_0^1 = 2 - 0.067(5) = 1.665$

$w_1^1 = 2 - 0.067(21) = 0.593$

Iteration - 2 $\eta_2 = 0.1 \, 5^{0.8} \therefore = 0.045$

$w_0^2 = 1.665 - 0.225 = 1.44$

$w_1^2 = 0.593 - 0.945 = -0.352$.

(d) · Learning rate comparision are following.

① . Constant Learning rate($\eta$) . may overshoot

② . Decaying Learning rate ($\eta$) . reduces . step size gradually.

③ . improves stability and smooth convergence

Hence Decaying($\eta$) · is more stable

(e) . Optimal . stepsize selection.

direction $d = -\nabla J \cdot (w)$

$\phi(\eta) = J(w + \eta d)$

Compare $\phi(m)$ and $\phi(m + 10^{-3})$

S-1 binary . search initial interval . $[0, 1] = [a, b]$

midpoint $m = \frac{a+b}{2}$.

Iteration - 1 $m = 0.5$.

Compare $\phi(0.5)$ . and $\phi(0.501)$

if $\phi(0.5) < \phi(0.501) \rightarrow$ interval $[0, 0.5]$

Iteration 2 $m = 0.25$, new interval $\boxed{[0, 0.25]}$ final reduced interval

S-2 Golden selection serch

initial . interval . $[0, 1]$. $m_1 = \frac{1}{4}$, $m_2 = \frac{3}{4}$

if $\phi(m_1) < \phi(m_2) \rightarrow$ interval becomes. $[0, \frac{3}{4}]$

If choose any two iterators points inside interval

$m_1 = 0.2$, $m_2 = 0.5$

Cont